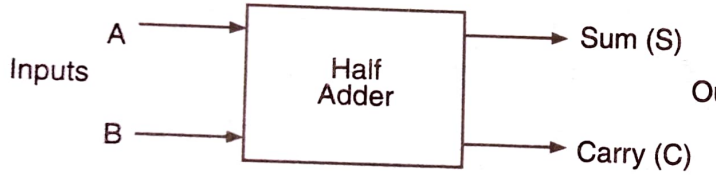
Half adder is a combinational logic circuit with two inputs and two outputs. It is the basic building block for addition of two single bit numbers. This circuit has two outputs namely carry and sum. Figure 3.29(a) shows the block diagram of half adder. An half adder circuit is designed to add two single bit binary numbers A and B. Hence, the truth table of a half adder will be as shown in figure 3.29(b).



(a) Block diagram

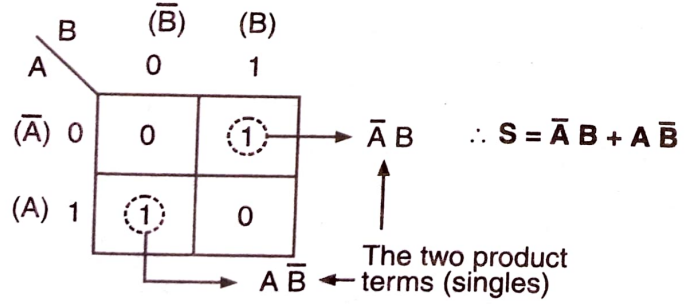
S. No.	Inputs		Outputs	
	A	B	Sum	Carry
1.	0	0	0	0
2.	0	1	1	0
3.	1	0	1	0
4.	1	1	0	1

(b) Truth table

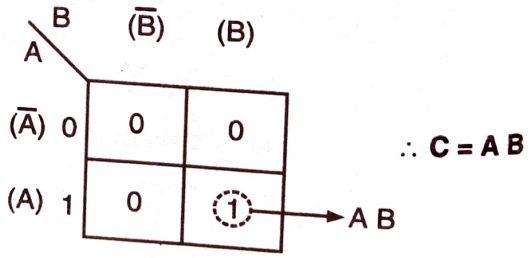
Fig. 3.29 An Half Adder

3.6.1. Karnaugh-Maps (K-Maps) and Simplified Expressions for Outputs

Figures 3.30 (a) and 3.30 (b) illustrates the K-maps for carry and sum outputs.



(a) K-map for sum output



(b) K-map for carry output

Fig. 3.30.

Boolean expressions for the sum (S) and carry (C) output may be obtained from the K-maps as under :

$$S = \bar{A}B + A\bar{B} = (A \oplus B) \quad \text{...(3.1)}$$

and $C = AB$

Therefore, figure 3.31 shows the half adder circuit.

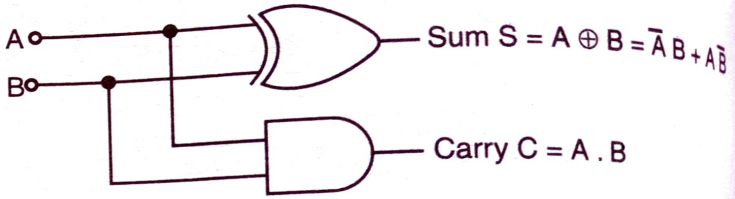


Fig. 3.31 An Half adder circuit

3.6.2. Drawback of Half Adder

Figure 3.32 shows the principle of adding two 2-bit numbers A and B.

Let Number $A = A_1 A_0$

and Number $B = B_1 B_0$

Then, the addition will take place as under

A_1 A_0 ← Number A

\+ B_1 B_0 ← Number B

\+ C_0 ← Carry generated from the addtion $(A_0 + B_0)$

C_1 S_1 S_0

Fig. 3.32. Two bit binary addition

A half adder can add A_0 and B_0 to produce S_0 and C_0. However, the addition of next bits requires the addition of A_1, B_1 and C_0. The addition of three bits is not possible to perform by using an half adder circuit. Therefore, practically we cannot use a half adder.

3.6.3. Half Adder Using Basic Gates

Let us consider equation (3.1) which states that,

Sum $\quad S = \bar{A}B + A\bar{B}$

and, Carry $\quad C = AB$

These equations can be implemented using the basic gates as depicted in figure 3.33.

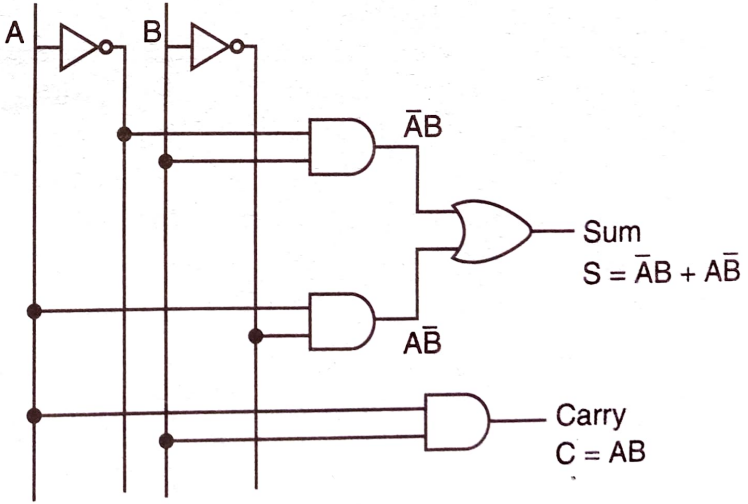


Fig. 3.33. An Half adder using basic gates

3.6.4. Half Adder Using Only NAND Gates

Again, equation (3.1) states that,

Sum, $$S = \bar{A}B + A\bar{B}$$

and Carry, $$C = AB$$

We consider the expressions separately, i.e.,

$$S = \bar{A}B + A\bar{B}$$

Now, let us take double inversion, so that, we have

$$S = \overline{\overline{\bar{A}B + A\bar{B}}}$$

Using the DeMorgan's theorem, we have

$$S = \overline{\overline{\bar{A}B} \cdot \overline{A\bar{B}}} \quad ...(3.2)$$

Similarly, we have $$C = AB = \overline{\overline{AB}} \quad ...(3.3)$$

Equations (3.2) and (3.3) can be implemented using only NAND gates as shown in figure 3.34.

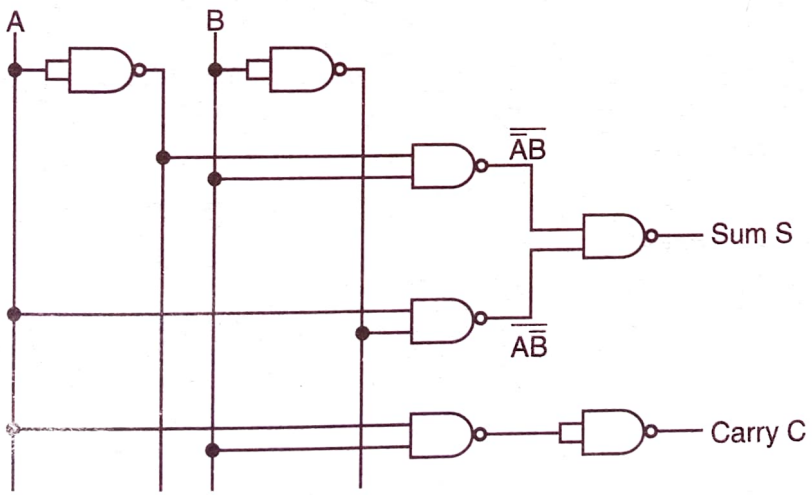


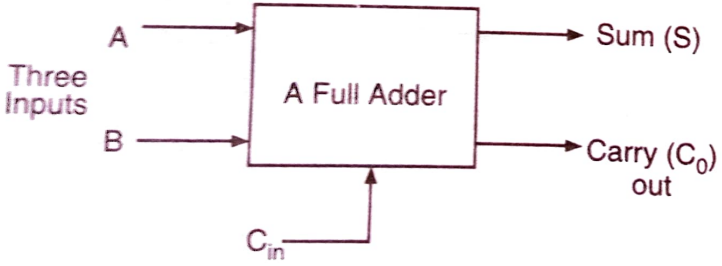
Fig. 3.34. An Half adder using only NAND gates

Natural ability without education has more often raised a man to glory and virtue than education without natural ability. — **M.T. Cicero**

3.7. A FULL ADDER

(RGTU, Bhopal, Sem. Exam, 2006-07)

To overcome the drawback of an Half Adder circuit, we develop a 3 single bit adder circuit called Full Adder. It can add two one-bit numbers A and B, and carry C_{in}. Basically, a full adder is a three input and two output combinational circuit. Figure 3.35 (a) depicts block diagram of a full adder and figure 3.35(b) illustrates its truth table.



(a) Block diagram

Inputs			Outputs	
A	B	C_{in}	S	C_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

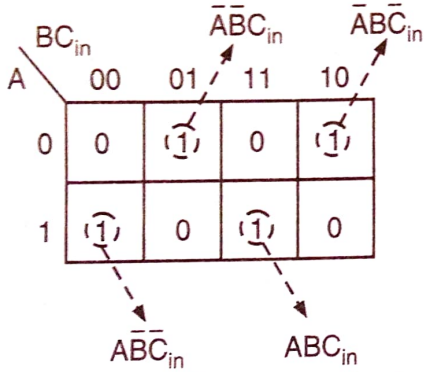
(b) Truth table

Fig. 3.35. A Full adder

3.7.1. K-maps for Full Adder

Figure 3.36 and figure 3.37 show the K-maps for the sum (S) and carry out (C_0) outputs and the corresponding Boolean expressions.

K-map for the sum output



Expression for sum output

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + ABC_{in} + A\bar{B}\bar{C}_{in}$$

$$S = C_{in}\underbrace{(\bar{A}\bar{B} + AB)}_{\text{EX-NOR}} + \bar{C}_{in}\underbrace{(\bar{A}B + A\bar{B})}_{\text{EX-OR}}$$

Therefore, $S = C_{in}(\bar{A}B + A\bar{B}) + \bar{C}_{in}(\bar{A}B + A\bar{B})$

Let $X = \bar{A}B + A\bar{B}$

Therefore, $S = C_{in}\bar{X} + \bar{C}_{in}X = C_{in} \oplus X$

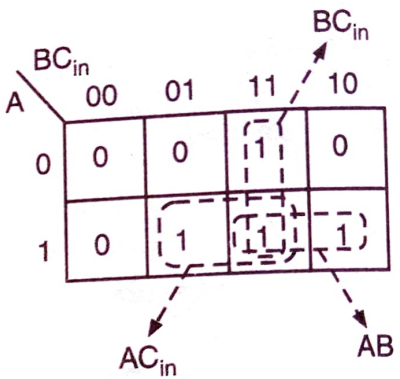
or $S = C_{in} \oplus (\bar{A}B + A\bar{B})$

But, $\bar{A}B + A\bar{B} = A \oplus B$

Therefore, $S = C_{in} \oplus A \oplus B$

Fig. 3.36. K-map and simplification for sum

K-map for C_0 output



Expression for C_0 output

$$C_0 = AB + AC_{in} + BC_{in}$$

Fig. 3.37. K-map and simplification for Carry out

DO YOU KNOW?

In digital systems, many devices must often share the same data path. This data path is often called a data bus.

To accomplish great things, we must not only act, but also dream: not only plan, but also believe.

— Anatole France

3.7.2. A Logic Diagram for Full Adder

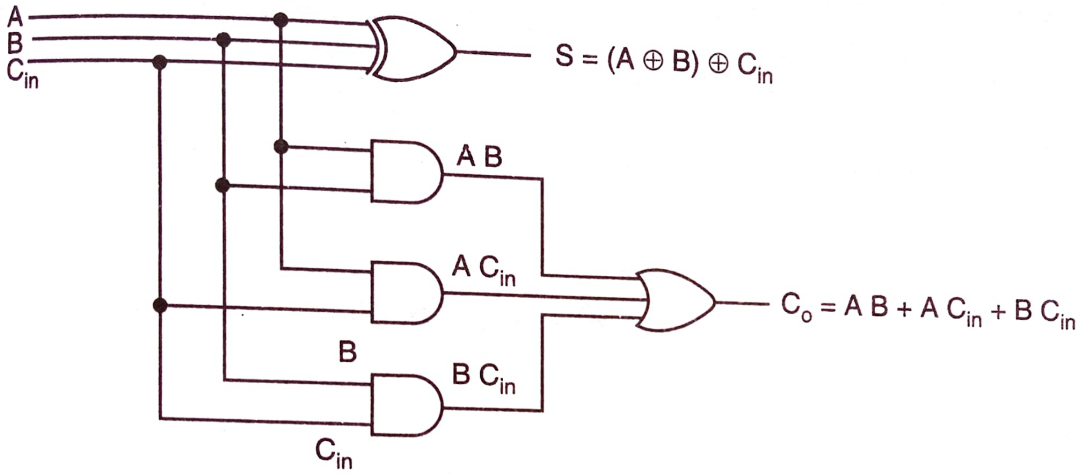


Fig. 3.38. A Full adder circuit

3.7.3. Construction of Full adder using Half Adder

A full adder circuit can also be constructed using two half adders as depicted in figure 3.39.

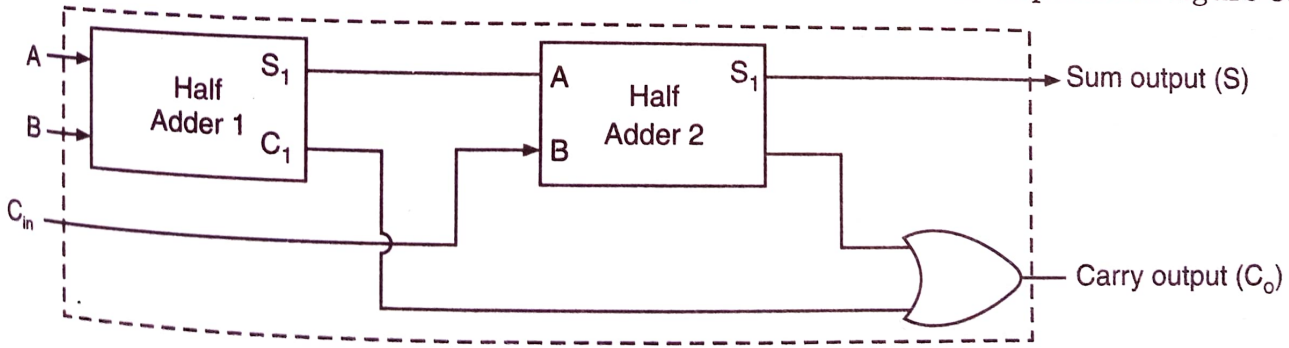


Fig. 3.39. Illustration of a Full adder using half adders

3.8. FULL ADDER USING HALF ADDERS (RGTU, Bhopal, Sem. Exam, 2007-08)

1. Circuit Diagram

Figure 3.40 shows that a full adder can be implemented using two half adders and an OR gate.

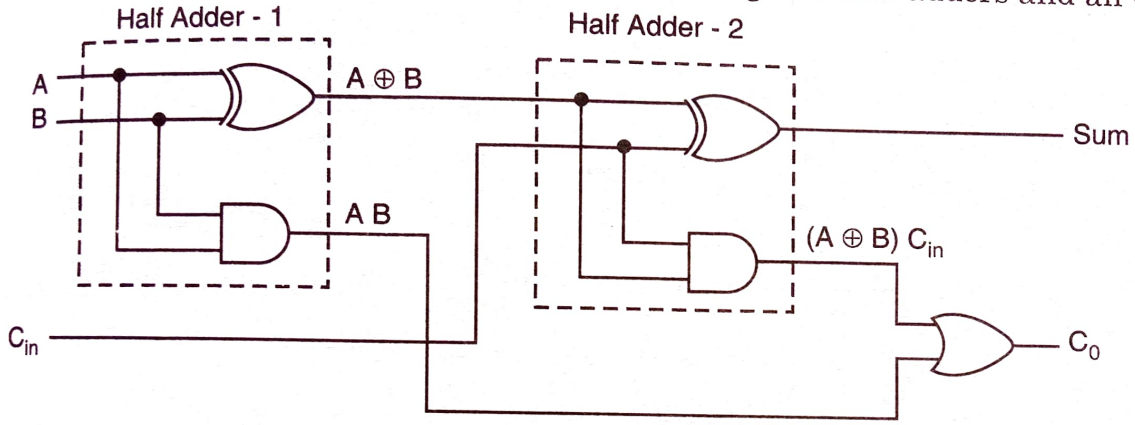


Fig. 3.40. Full adder using two half adders

Now, let us illustrate that the circut in figure 3.40 acts as a full adder.

2. Proof

Let us consider figure 3.40 and write the expression for sum output as under:

$$S = (A \oplus B) \oplus C_{in} = A \oplus B \oplus C_{in}$$

This expression is same as that obtained for a full adder.

Now, we write the expression for carry output C_0 as under:

$$C_0 = (A \oplus B)\, C_{in} + AB = (\bar{A}B + A\bar{B})\, C_{in} + AB$$

1. Definition

Half subtractor may be defined as a combinational circuit with two inputs and two outputs (i.e., difference and borrow). A half subtractor produces the difference between the two binary bits at the input and also produces an output (i.e., borrow) to indicate if a 1 has been borrowed. In subtraction (A – B), A is called as minuend bit and B is called as subtrahend bit.

2. Truth Table

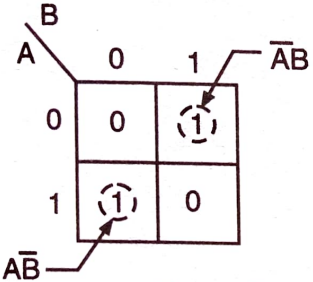
Table 3.2 illustrates a truth table showing the outputs of a half subtractor for all the possible combinations of input.

Table 3.2. Truth table for an half subtractor

	Inputs		Outputs	
S. No.	**A**	**B**	**Difference D (A – B)**	**Borrow B_0**
1.	0	0	0	0
2.	0	1	1	1
3.	1	0	1	0
3.	1	1	0	0

3. K-maps for Difference and Borrow Outputs

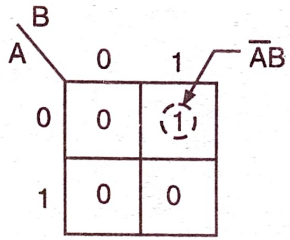
Figure 3.41 shows the K-maps for the two outputs of a half subtractor.



Therefore, difference $D = \bar{A}B + A\bar{B}$

Hence, $D = A \oplus B$

(a) K-map and simplification for difference output



Therefore, borrow $B_0 = \bar{A}B$

(b) K-map and simplification for the borrow output

Fig. 3.41.

4. Logic Diagram

Figure 3.41 (c) shows a logic diagram for an half subtractor.

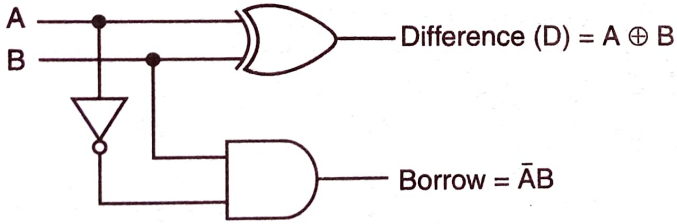


Fig. 3.41. (c) An Half subtractor circuit

5. Drawback of An Half subtractor

An half subtractor can only perform the subtraction of two binary bits. However, while performing the subtraction, it does not take into account the borrow of the lower significant stage.

6. An Half subtractor Using Basic Gates

The expressions for the difference and borrow outputs may be listed as under:

$$D = \bar{A}B + A\bar{B} \text{ and } B_0 = \bar{A}B$$

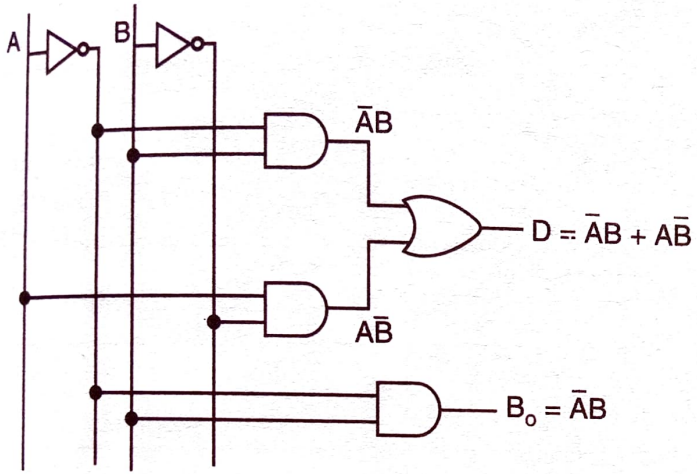


Fig. 3.42. An Half subtractor using basic gates

DO YOU KNOW?

Even though many devices can be *riding* on the bus, there can be only one bus *driver* at any one time. Thus, devices must take turns applying logic signals to the data bus.

7. An Half subtractor Using NAND Gates only

The expression for difference output is given by

$$D = \bar{A}B + A\bar{B}$$

Now, taking the double inversion, we obtain

$$D = \overline{\overline{\bar{A}B + A\bar{B}}}$$

Applying the DeMorgan's theorem, we obtain

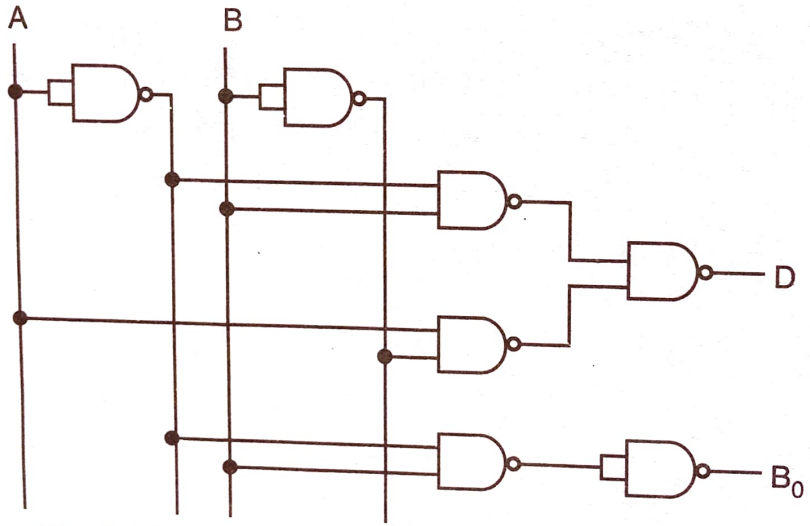


Fig. 3.43. An Half subtractor using NAND gates only

$$D = \overline{\overline{\bar{A}B}.\overline{A\bar{B}}} \qquad \ldots(3.4)$$

Similarly, the other output B_0 will be given by,

$$B_0 = \bar{A}B = \overline{\overline{\bar{A}B}} \qquad \ldots(3.5)$$

Equations (3.4) and (3.5) can be implemented as depicted in figure 3.43.

3.11. A FULL SUBTRACTOR

(RTU, Kota, Sem. Exam., 2006-07) (10 marks)

1. Definition

The drawback of an half subtractor can be overcome by a full subtractor. Basically, a full subtractor is a combinational circuit with three inputs A, B and B_{in} and two outputs D and B_0. Here, A is the minuend, B is subtrahend, B_{in} is the borrow produced by the previous stage, D is the difference output and B_0 is the borrow output.

2. Truth Table for a Full subtractor

Table 3.3 represents a truth table for a full subtractor.

Table 3.3. Truth table for a full subtractor

	Inputs			Outputs	
S. No.	A (Minuend)	B (subtrahend)	B_{in} (Previous borrow)	$(A - B - B_{in})$	B_0
1.	0	0	0	0	0
2.	0	0	1	1	1
3.	0	1	0	1	1
4.	0	1	1	0	1
5.	1	0	0	1	0
6.	1	0	1	0	0
7.	1	1	0	0	0
8.	1	1	1	1	1

3. K-maps and Simplifications

Figures 3.44 (a) and 3.44 (b) show K-maps for D and B_0 outputs.

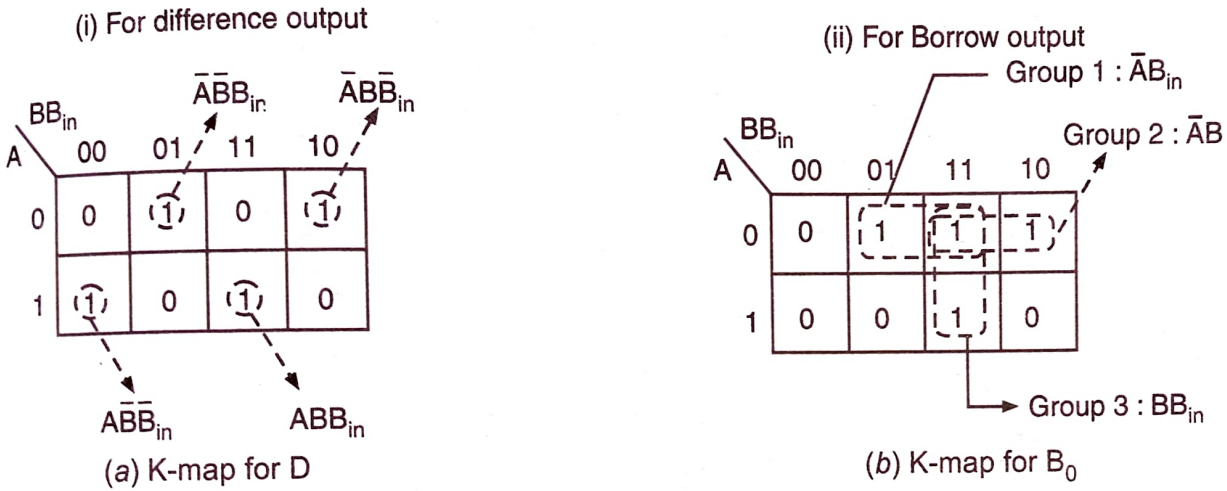


Fig. 3.44.

Therefore, $D = \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + ABB_{in}$

Hence, $B_0 = \bar{A}B_{in} + \bar{A}B + BB_{in}$

4. Simplification for difference output

From figure 3.44 (a), we have, $D = \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + AB\,B_{in}$

$$= B_{in}\underbrace{(\bar{A}\bar{B} + AB)}_{\text{EX-NOR}} + \bar{B}_{in}\underbrace{(\bar{A}B + A\bar{B})}_{\text{EX-OR}}$$

Therefore, we have $D = B_{in}\overline{(A \oplus B)} + \bar{B}_{in}(A \oplus B)$

Let $A \oplus B = C,$

Hence, we write $D = B_{in}\,\bar{C} + \bar{B}_{in}C = B_{in} \oplus C$

Therefore, $D = B_{in} \oplus A \oplus B$

5. Simplification for Borrow output

From figure 3.44 (b), we have

$$B_0 = \bar{A}B_{in} + \bar{A}B + BB_{in}$$

Here, no further simplification is possible.

6. Logic

Here,

6. Logic diagram for a full subtractor

Figure 3.45 illustrates the logic diagram for a full subtractor.

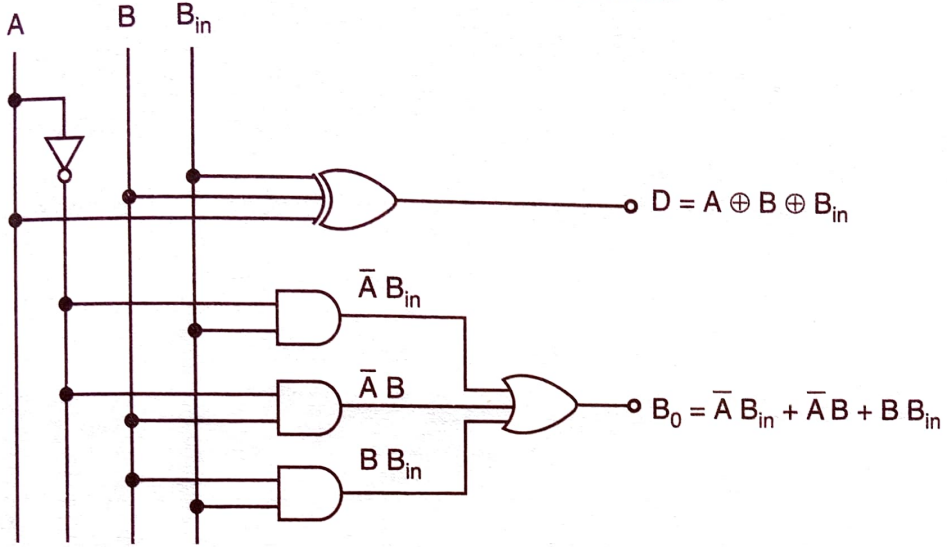


Fig. 3.45. Illustration of Logic diagram for a full subtractor

3.11.1. Construction of a Full Subtractor using Half Subtractors

1. Logic Diagram

Figure 3.46 depicts the implementation of a full subtractor using two half subtractors and an OR gate.

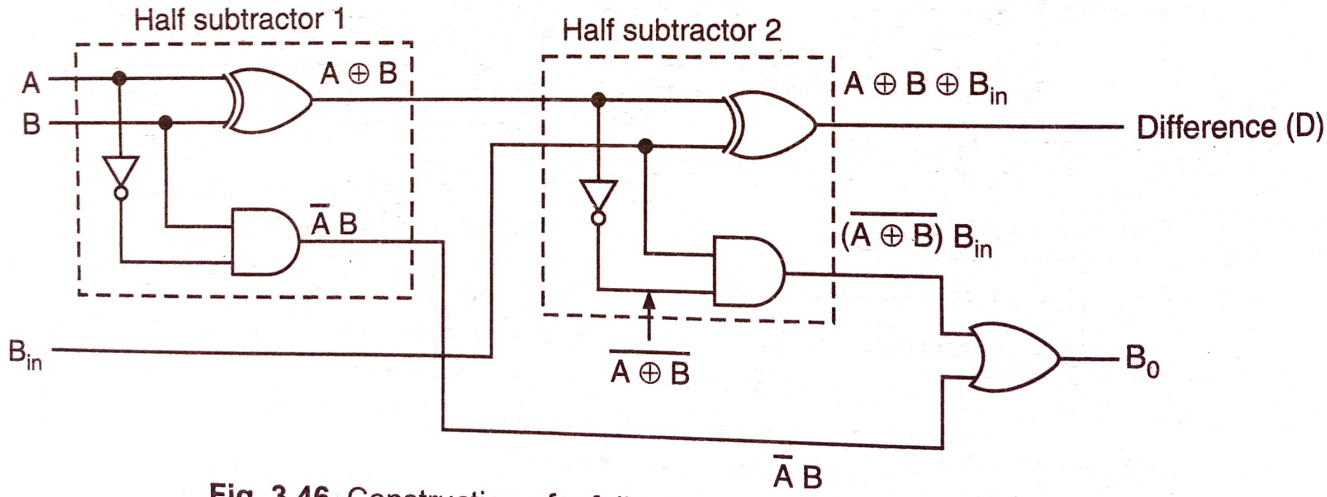


Fig. 3.46. Construction of a full subtractor using half subtractors

Let us illustrate that the circuit